Data chapter no 1 Introduction

The information collected with experiments and surveys is called data. Data is actual numerical values that variables assume.

Datum : A single numerical fact is called a datum.

POPULATION: A group of all possible objects or individuals about some characteristic of interest is called population. For example population of students in a college, population of patients in a hospital.

SAMPLE:

A small part of the population selected to represent population is called sample.

MEANINGS OF STATISTICS

The words statistics has 3 different meanings

1-STATISTICS IN PLURAL SENSE:

In plural sense statistics refers to aggregate of facts or numbers in any field of study. For example statistics of births, price statistics etc.

2- STATISTICS IN SINGULAR SENSE:

In singular sense it is a science consisting of methods used in collection, presentation, analysis and interpretation of numerical data.

3- STATISTICS AS PLURAL OF STATISTIC:

In this sense statistics is used as plural of statistic. By statistic we mean numerical value calculated from sample data.

PARAMETER:

A numerical meaure that describes the characteristic of a population is called population parameter. Parameter is denoted by Greek letters. It is a constant.

STATISTIC:

A numerical measure that describes the characteristic of a sample is called sample statistic. Statistic is denoted by Roman letters. It is a variable.

BRANCHES OF STATISTICS:

Statistics is divided into two branches.

1- DESCRIPTIVE STATISTICS:

Descriptive statistics consists of methods and techniques used in collection summarization, presentation and analysis of data. This area of study includes graphical display of data and computation of some numerical quantities that provide information about centre of data and dispersion of observations.

2- INFERENTIAL STATISTICS:

Inferential statistics consists of methods used to draw conclusions about the characteristic of the population on the basis of information contained in the sample. This area of study includes estimation of population parameter and testing of statistical hypothesis. It is based on probability theory.

FUNCTIONS OR USES OF STATISTICS:

1- Statistics presents facts in a form which is easily understood.

2- Numerical facts give more precise information than facts expressed in general terms.

3- It facilitates comparison.

4- It helps in prediction

5- It helps in the formation of suitable policies

LIMITATIONS OF STATISTICS:

1- Statistical results are true only on the average or in the long run.

2- Statistics deals with facts which can be numerically measured.

3- Use of statistical techniques demands great expertise and experience

4- Statistical results sometimes lead to confusion

5- Statistical laws and rules do not hold good in every case.

VARIABLE:

A characteristic that varies from one individual or object to another is called a variable. For example heights of students, number of children per family, eye color, education level etc.

Variables are of two types.

1- QUANTITATIVE VARIABLE:

A characteristic that varies only in quantity from one individual to another is called quantitative variable. For example heights of students, speed of car, shoe sizes etc.

Quantitative variables are further subdivided into two types.

i) DISCRETE VARIABLE:

A variable which can assume only some specific values within a given range is called discrete variable. For example number of students in the class, shoe size etc. These are counts.

ii) CONTINUOUS VARIABLE:

A variable which can assume any possible value within a given range is called a continuous variable. For example height, weight, temperature etc. these are measurements

2- QUALITATIVE VARIABLE:

A characteristic that varies only in quality from one individual or object to another is called qualitative. For example sex, religion, eye color etc.It is also known as attribute

CONSTANTS

An observed characteristic is called constant if it attains identical values from person to person, place to place or time to time.

QUANTITATIVE DATA:

Any data observed in numerical form is called quantitative data.

QUALITAIVER DATA:

Any data observed in non numerical form is called qualitative data.

DISCRETE DATA:

Any data observed in counting style numerals is called discrete data.

CONTINUOUS DATA:

Any data observed in measuring style numerals is called continuous data.

PRIMARY DATA:

The data that have been collected and have not undergone any statistical treatment are called primary data

SECONDARY DATA:

The data that have undergone some statistical treatment at least once are called secondary data.

SOURCES OF SECONDARY DATA.

The secondary data may be taken from the following sources (i) Published sources (ii) unpublished sources.

METHODS OF COLLECTING PRIMARY DATA

i. Direct personal observation
ii. Indirect oral investigation
iii. Through questionnaire
iv. Registration

METHODS OF COLLECTING SECONDARY DATA:

I. Official sources
II. Semi-official sources
III. Private sources
IV. Research organizations

Impt.

Measurement Scales

There are four types of measurement scales.

1. Nominal
2. Ordinal
3. Interval
4. Ratio

Nominal Scale

For the nominal level of measurement observations of a qualitative variable can only be classified and counted. There is no particular order to the labels.

Examples of nominal variables: Religion, Gender, Country

For the nominal level, the only measurement involved consists of counts. Sometimes we convert these counts to percentages.

The nominal level has the following properties

1. The variable of interest is divided into categories or outcomes
2. There is no natural order to the outcome.

Ordinal Scale

These scales allow for categorization as in a nominal scale, but they allow for ranking.

For example :our preferences for linking different cities,Linking of different products

Even though differences in ranking of objects, persons or events investigated are clearly known but we do not know their magnitude

Permitted statistics Frequencies, Median, Mode

Interval Scale

Interval scale contains information available in ordinal scale but with added benefit of the magnitude of ranking. Interval scales have equal distances between the points of a scale.

For Example : Our opinion about the quality of education in scaling in college

1) Strongly disagree 2) Somewhat disagree 3) Neither

4) Somehow agree 5)Strongly agree

It is more powerful scale then nominal and ordinal scales.

Ratio Scale

Ratio scale is the most comprehensive scale and it overcomes the disadvantage of the arbitrary origin point of the interval scale. It has all the characteristics of other three measurement scales with the additional benefits of an absolute, meaningful zero point.

Example: Weight, Age, Income, Profits, Expenses etc

Time Series Data:

An arrangement of data by successive time periods is called times series data OR

Set of Observations recorded according to the time of their occurrence is called time series Data.

For example automobile sales figures, stock prices, gross domestic product, etc.

Cross-sectional data:

The data which is collected on one or more variables at a single period of time.

A cross-sectional data-set consists of a sample of individuals , households etc taken at a given point in time.

Panel data(Pooled Data):

These are models that combine Time series data and cross-section data.

In panel data the same cross-sectional unit is surveyed over time, so we have data which is pooled over space as well as time.

It has dimensions of both time series and cross-section.

For Example: suppose we have wage, education, and employment history for a set of individuals follow over a ten-year period